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NUMERICAL ANALYSIS

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Numerical Analysis

9 Method of Least Squares

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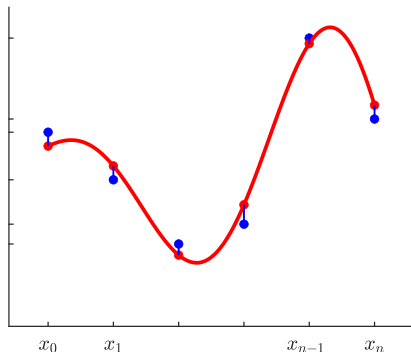


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Suppose the formula of the function g depends on some parameters $\mathbf{a}$:

$$g(x; \mathbf{a}).$$

Suppose we have measurements y_i ($i = 0, 1, \dots, n$) of the function values at the mesh points x_i . Our goal is to find the parameter values for which the corresponding function g differs from the measurements with the “smallest error”. This problem is called **curve fitting**.



Possible error formulas:

$$F_1(\mathbf{a}) := \max\{|g(x_i; \mathbf{a}) - y_i| : i = 0, 1, \dots, n\}$$

or

$$F_2(\mathbf{a}) := \sum_{i=0}^n |g(x_i; \mathbf{a}) - y_i|.$$

The so-called **least square error**:

$$F(\mathbf{a}) := \sum_{i=0}^n (g(x_i; \mathbf{a}) - y_i)^2.$$

Let $\bar{\mathbf{a}}$ be the minimum point of $F(\mathbf{a})$, and consider the graph of the function $g(x; \bar{\mathbf{a}})$ corresponding to $\bar{\mathbf{a}}$ as the best fitted curve to the data points. This is called the **method of least squares**.

9.1 Line Fitting

Given data points (x_i, y_i) , $i = 0, 1, \dots, n$, where at least some of the mesh points x_i are different. We are looking for a linear function of the form

$$g(x) = ax + b$$

which minimizes the least square error

$$F(a, b) := \sum_{i=0}^n (ax_i + b - y_i)^2. \quad (1)$$

The function F is continuously partially differentiable with respect to a and b , and

$$\begin{aligned} \frac{\partial F}{\partial a}(a, b) &= 2 \sum_{i=0}^n (ax_i + b - y_i)x_i, \\ \frac{\partial F}{\partial b}(a, b) &= 2 \sum_{i=0}^n (ax_i + b - y_i). \end{aligned} \quad (2)$$

Making the partial derivatives in (2) equal to 0, and rearranging the system we get the so-called **Gaussian normal equations**:

$$\begin{aligned} a \sum_{i=0}^n x_i^2 + b \sum_{i=0}^n x_i &= \sum_{i=0}^n x_i y_i, \\ a \sum_{i=0}^n x_i + b(n+1) &= \sum_{i=0}^n y_i. \end{aligned} \tag{3}$$

This system is solvable if the determinant of its coefficient matrix

$$d := \det \begin{pmatrix} \sum_{i=0}^n x_i^2 & \sum_{i=0}^n x_i \\ \sum_{i=0}^n x_i & n+1 \end{pmatrix} = (n+1) \sum_{i=0}^n x_i^2 - \left(\sum_{i=0}^n x_i \right)^2$$

is nonzero.

The Cauchy–Bunyakovszky–Schwarz inequality yields

$$\left(\sum_{i=0}^n x_i\right)^2 = \left(\sum_{i=0}^n 1 \cdot x_i\right)^2 \leq \sum_{i=0}^n 1 \sum_{i=0}^n x_i^2 = (n+1) \sum_{i=0}^n x_i^2,$$

therefore $d \geq 0$ holds. If we assume that there are at least two distinct mesh points x_i , then the strict inequality $d > 0$ holds. Hence system (3) has a unique solution which can be given in the following form:

$$\begin{aligned}\bar{a} &= \frac{(n+1) \left(\sum_{i=0}^n x_i y_i\right) - \left(\sum_{i=0}^n x_i\right) \left(\sum_{i=0}^n y_i\right)}{(n+1) \left(\sum_{i=0}^n x_i^2\right) - \left(\sum_{i=0}^n x_i\right)^2}, \\ \bar{b} &= \frac{\left(\sum_{i=0}^n x_i^2\right) \left(\sum_{i=0}^n y_i\right) - \left(\sum_{i=0}^n x_i y_i\right) \left(\sum_{i=0}^n x_i\right)}{(n+1) \left(\sum_{i=0}^n x_i^2\right) - \left(\sum_{i=0}^n x_i\right)^2}.\end{aligned}$$

Function F has a local extremum at $(\bar{a}, \bar{b})$ if

$$D(\bar{a}, \bar{b}) := \frac{\partial^2 F}{\partial a^2}(\bar{a}, \bar{b}) \cdot \frac{\partial^2 F}{\partial b^2}(\bar{a}, \bar{b}) - \left(\frac{\partial^2 F}{\partial a \partial b}(\bar{a}, \bar{b}) \right)^2 > 0.$$

It is easy to compute that

$$\frac{\partial^2 F}{\partial a^2}(\bar{a}, \bar{b}) = 2 \sum_{i=0}^n x_i^2, \quad \frac{\partial^2 F}{\partial b^2}(\bar{a}, \bar{b}) = 2(n+1), \quad \frac{\partial^2 F}{\partial a \partial b}(\bar{a}, \bar{b}) = 2 \sum_{i=0}^n x_i.$$

Hence

$$D(\bar{a}, \bar{b}) = 4(n+1) \sum_{i=0}^n x_i^2 - 4 \left(\sum_{i=0}^n x_i \right)^2 = 4d,$$

which we know that it is positive. Since $\frac{\partial^2 F}{\partial a^2}(\bar{a}, \bar{b}) > 0$, F has a local minimum at $(\bar{a}, \bar{b})$, and hence it is also a global minimum.

Theorem:

Given data points (x_i, y_i) ($i = 0, 1, \dots, n$) such that there exist i and j with $x_i \neq x_j$. Then the problem

$$\min_{(a,b) \in \mathbb{R}^2} \sum_{i=0}^n (ax_i + b - y_i)^2$$

has a unique solution, which satisfies the Gaussian normal equations (3).

Example

Given the following data:

x_i	0	1.0	1.5	3.0	3.5	4.0	4.5	6.0
y_i	1.2	1.4	1.7	2.4	2.7	3.1	3.1	4.1

Find a line of best fit to the data points. We fill out the next table:

x_i	y_i	x_i^2	$x_i y_i$
0.0	1.2	0.00	0.00
1.0	1.4	1.00	1.40
1.5	1.7	2.25	2.55
3.0	2.4	9.00	7.20
3.5	2.7	12.25	9.45
4.0	3.1	16.00	12.40
4.5	3.1	20.25	13.95
6.0	4.1	36.00	24.60
23.5	19.7	96.75	71.55

Example cont.

Substituting the sums to the Gaussian normal equations

$$a \sum_{i=0}^n x_i^2 + b \sum_{i=0}^n x_i = \sum_{i=0}^n x_i y_i,$$

$$a \sum_{i=0}^n x_i + b(n+1) = \sum_{i=0}^n y_i$$

we get

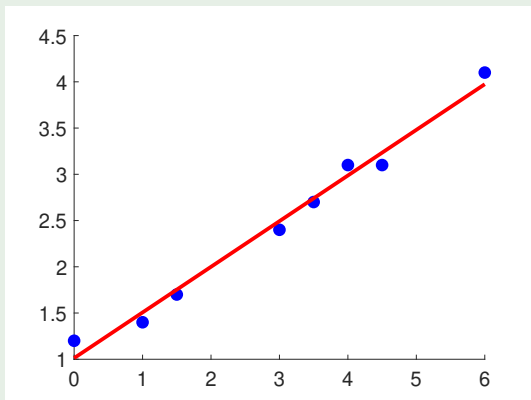
$$96.75a + 23.5b = 71.55$$

$$23.5a + 8b = 19.70.$$

Its solution is $a = 0.49357$ and $b = 1.01263$. The graph of the corresponding line $y = 0.49357x + 1.01263$ can be seen in the next figure. The error of the fitting is

$$\sum_{i=0}^7 (0.49357x_i + 1.01263 - y_i)^2 = 0.10604.$$

Example cont.



Line fitting: $y = 0.49357x + 1.01263$

9.2 Polynomial Curve Fitting

Next we study the problem of **polynomial curve fitting**. Given data points (x_i, y_i) ($i = 0, 1, \dots, n$). We find a polynomial of degree m of best fit to the data points, i.e., we are looking for parameters $a_m, a_{m-1}, \dots, a_0$ which minimize the least square error function

$$F(a_m, a_{m-1}, \dots, a_1, a_0) := \sum_{i=0}^n (a_m x_i^m + a_{m-1} x_i^{m-1} + \dots + a_1 x_i + a_0 - y_i)^2,$$

a function of $m + 1$ variables.

If $n \leq m$, then there is a polynomial of degree m which interpolates the given data (the minimal value of F is 0). So the coefficients can be obtained by polynomial interpolation.

Therefore we assume for the rest of this section that $m < n$, and in this case F can be positive at every point.

We get that F can have an extremum at a point where all partial derivatives are equal to 0.

$$\begin{aligned}\frac{\partial F}{\partial a_m}(a_m, a_{m-1}, \dots, a_0) &= 2 \sum_{i=0}^m (a_m x_i^m + a_{m-1} x_i^{m-1} + \dots + a_0 - y_i) x_i^m, \\ \frac{\partial F}{\partial a_{m-1}}(a_m, a_{m-1}, \dots, a_0) &= 2 \sum_{i=0}^m (a_m x_i^m + a_{m-1} x_i^{m-1} + \dots + a_0 - y_i) x_i^{m-1}, \\ &\vdots \\ \frac{\partial F}{\partial a_0}(a_m, a_{m-1}, \dots, a_0) &= 2 \sum_{i=0}^m (a_m x_i^m + a_{m-1} x_i^{m-1} + \dots + a_0 - y_i).\end{aligned}$$

Making the partial derivatives equal to 0 and rearranging the resulting system, we get the **normal equations**

$$\begin{array}{ccccccc}
 a_m \sum_{i=0}^n x_i^{2m} & + a_{m-1} \sum_{i=0}^n x_i^{2m-1} & + \cdots & + a_1 \sum_{i=0}^n x_i^{m+1} & + a_0 \sum_{i=0}^n x_i^m & = \sum_{i=0}^n x_i^m y_i \\
 a_m \sum_{i=0}^n x_i^{2m-1} & + a_{m-1} \sum_{i=0}^n x_i^{2m-2} & + \cdots & + a_1 \sum_{i=0}^n x_i^m & + a_0 \sum_{i=0}^n x_i^{m-1} & = \sum_{i=0}^n x_i^{m-1} y_i \\
 \vdots & & & & & \vdots \\
 a_m \sum_{i=0}^n x_i^{m+1} & + a_{m-1} \sum_{i=0}^n x_i^m & + \cdots & + a_1 \sum_{i=0}^n x_i^2 & + a_0 \sum_{i=0}^n x_i & = \sum_{i=0}^n x_i y_i \\
 a_m \sum_{i=0}^n x_i^m & + a_{m-1} \sum_{i=0}^n x_i^{m-1} & + \cdots & + a_1 \sum_{i=0}^n x_i & + a_0(n+1) & = \sum_{i=0}^n y_i
 \end{array}$$

We prove that the linear system (4) has a unique solution. For this it is enough to show that the coefficient matrix

$$\mathbf{A} := \begin{pmatrix} \sum_{i=0}^n x_i^{2m} & \sum_{i=0}^n x_i^{2m-1} & \cdots & \sum_{i=0}^n x_i^{m+1} & \sum_{i=0}^n x_i^m \\ \sum_{i=0}^n x_i^{2m-1} & \sum_{i=0}^n x_i^{2m-2} & \cdots & \sum_{i=0}^n x_i^m & \sum_{i=0}^n x_i^{m-1} \\ \vdots & \vdots & & \vdots & \vdots \\ \sum_{i=0}^n x_i^m & \sum_{i=0}^n x_i^{m-1} & \cdots & \sum_{i=0}^n x_i & \sum_{i=0}^n 1 \end{pmatrix}$$

is invertible. It is enough to show that $\mathbf{A}$ is positive definite. The jk -th element of the matrix $\mathbf{A}$ is given by formula

$$\sum_{i=0}^n x_i^{2m+2-j-k}, \quad j, k = 1, 2, \dots, m+1.$$

Let $\mathbf{z} = (z_1, z_2, \dots, z_{m+1}) \in \mathbb{R}^{m+1}$. Simple calculations give

$$\begin{aligned}\mathbf{z}^T \mathbf{A} \mathbf{z} &= \sum_{j=1}^{m+1} \sum_{k=1}^{m+1} \sum_{i=0}^n x_i^{2m+2-j-k} z_j z_k \\ &= \sum_{i=0}^n \sum_{j=1}^{m+1} \sum_{k=1}^{m+1} x_i^{m+1-j} z_j x_i^{m+1-k} z_k \\ &= \sum_{i=0}^n \left(\sum_{j=1}^{m+1} x_i^{m+1-j} z_j \right)^2.\end{aligned}$$

Suppose that

$$\mathbf{z}^T \mathbf{A} \mathbf{z} = 0.$$

Then we have that

$$\sum_{j=1}^{m+1} x_i^{m+1-j} z_j = 0, \quad i = 0, 1, \dots, n.$$

So if there are $m + 1$ distinct mesh points, then the polynomial

$$p(x) := \sum_{j=1}^{m+1} z_j x^{m+1-j}$$

of degree at most m has $m + 1$ distinct roots. Therefore the Fundamental theorem of algebra yields that p must be identically equal to 0, i.e.,

$$z_j = 0, \quad j = 1, 2, \dots, m + 1.$$

Hence we get that $\mathbf{A}$ is positive definite, and so system (4) has a unique solution denoted by $\bar{\mathbf{a}}$.

Since

$$\frac{\partial^2 F}{\partial a_j \partial a_k}(\bar{\mathbf{a}}) = 2 \sum_{i=0}^n x_i^{j+k},$$

we get

$$F''(\bar{\mathbf{a}}) = 2\mathbf{A}.$$

Therefore F has a local minimum at $\bar{\mathbf{a}}$, and since F is a quadratic function, it is also a global minimum. We can summarize our result in the next theorem.

Theorem:

Let $m < n$, and given data point (x_i, y_i) ($i = 0, 1, \dots, n$) such that there exist at least $m + 1$ distinct mesh points x_i . Then the problem

$$\min_{(a_m, \dots, a_0) \in \mathbb{R}^{m+1}} \sum_{i=0}^n (a_m x_i^m + a_{m-1} x_i^{m-1} + \dots + a_1 x_i + a_0 - y_i)^2$$

has a unique solution which satisfies the normal equations (4).

Example

Find a parabola of best fit to the data

x_i	-1.0	0.0	0.5	1.0	2.0	2.5	3.0
y_i	1.4	1.9	1.6	1.7	0.2	-0.1	-2.0

We fill out the table:

x_i	y_i	x_i^4	x_i^3	x_i^2	$x_i^2 y_i$	$x_i y_i$
-1.0	1.4	1.0000	-1.000	1.00	1.400	-1.40
0.0	1.9	0.0000	0.000	0.00	0.000	0.00
0.5	1.6	0.0625	0.125	0.25	0.400	0.80
1.0	1.7	1.0000	1.000	1.00	1.700	1.70
2.0	0.2	16.0000	8.000	4.00	0.800	0.40
2.5	-0.1	39.0625	15.625	6.25	-0.625	-0.25
3.0	-2.0	81.0000	27.000	9.00	-18.000	-6.00
8.0	4.7	138.1250	50.750	21.50	-14.325	-4.75

Example cont.

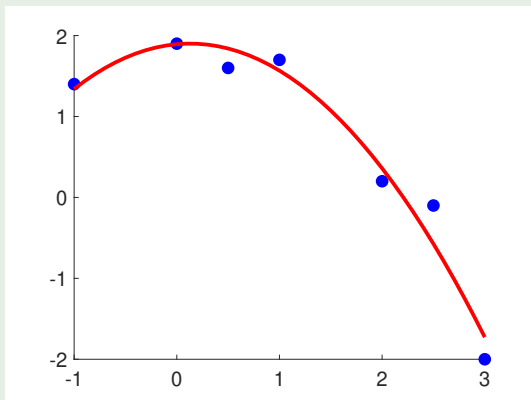
In the last line we compute the sum of the numbers in the respective columns, and we use these numbers in the normal equations (4):

$$\begin{aligned} 138.125a + 50.75b + 21.5c &= -14.325 \\ 50.75a + 21.5b + 8c &= -4.75 \\ 21.5a + 8b + 7c &= 4.7. \end{aligned}$$

Its solution is $a = -0.440076$, $b = 0.113226$ and $c = 1.89369$. The graph of the corresponding parabola and the given data point can be seen in the next figure. The error of the fitting is

$$\sum_{i=0}^6 (-0.44008x_i^2 + 0.11323x_i + 1.89369 - y_i)^2 = 0.40338.$$

Example cont.



Parabola fitting: $y = -0.44008x^2 + 0.11323x + 1.89369$

9.3 Special Nonlinear Curve Fitting

Consider an example for nonlinear curve fitting. Suppose we would like to fit an exponential function of the form be^{ax} to given data (x_i, y_i) ($i = 0, 1, \dots, n$). The least square error in this case will define the function

$$F(a, b) = \sum_{i=0}^n (be^{ax_i} - y_i)^2,$$

whose critical points are the solutions of the nonlinear system

$$\begin{aligned} 2 \sum_{i=0}^n (be^{ax_i} - y_i) be^{ax_i} x_i &= 0 \\ 2 \sum_{i=0}^n (be^{ax_i} - y_i) e^{ax_i} &= 0. \end{aligned}$$

We cannot solve this system analytically, and it is not easy to analyse whether this system has a unique solution or more, or in the latter case, which solution minimizes the error function. Certainly, we can solve the system numerically, or we can minimize F by a numerical method.

Use the method of **linearization** for this special example. Take the natural logarithm of both sides of the equation

$$y = be^{ax},$$

then we get the relation

$$\ln y = \ln b + ax.$$

We introduce the new variables:

$$X := x, \quad Y := \ln y, \quad A := a \quad \text{and} \quad B := \ln b.$$

So we can fit a line of the form

$$Y = AX + B \quad \text{to the data points} \quad (x_i, \ln y_i).$$

Let $\bar{A}$ and $\bar{B}$ be the solution of this linear fitting. Then the function $\bar{b}e^{\bar{a}x}$ can be considered as the best fit to the points (x_i, y_i) , where

$$\bar{a} = \bar{A}, \quad \bar{b} = e^{\bar{B}}.$$

Example

Fit an exponential function be^{ax} to the data

x_i	0.0	1.0	1.5	2.0	3.0	4.0
y_i	0.3	0.7	0.9	1.2	1.8	2.7

using linearization. The linearized data can be seen in the next table:

x_i	y_i	$\ln y_i$	x_i^2	$x_i \ln y_i$
0.0	0.3	-1.203973	0.00	0.000000
1.0	0.7	-0.356675	1.00	-0.356675
1.5	0.9	-0.105361	2.25	-0.158041
2.0	1.2	0.182322	4.00	0.364643
3.0	1.8	0.587787	9.00	1.763360
4.0	2.7	0.993252	16.00	3.973007
11.5		0.097352	32.25	5.586294

Example cont.

The corresponding Gaussian normal equations are

$$32.25A + 11.5B = 5.586294$$

$$11.5A + 6B = 0.097352,$$

which give $A = 0.528951$ and $B = -0.997597$. So the solution of the linearized fitting is

$$y = 0.368765e^{0.528951x}.$$

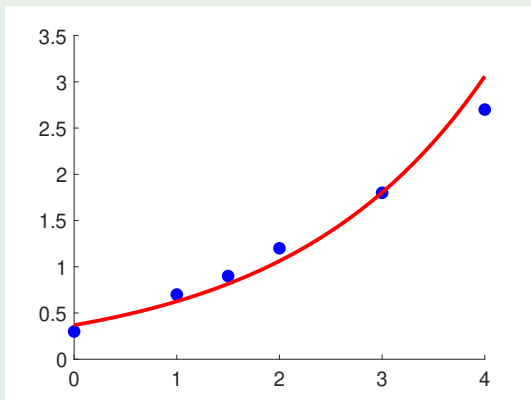
Its graph and the data points can be seen in the figure below. The error of the linear fitting is

$$\sum_{i=0}^5 (0.528951x_i - 0.997597 - \ln y_i)^2 = 0.095396,$$

and the error of the nonlinear fitting is

$$\sum_{i=0}^5 (0.368765e^{0.528951x_i} - y_i)^2 = 0.165543.$$

Example cont.



Fitting an exponential function: $0.368765e^{0.528951x}$

Example

Fit a power function of the form bx^a to the given data

x_i	0.5	1.0	1.5	2.5	3.0
y_i	0.7	1.1	1.6	2.1	2.3

From the equation $y = bx^a$ using the method of linearization we get

$$\ln y = a \ln x + \ln b.$$

Then $\ln y$ depends linearly on $\ln x$. We therefore fit a line to the data points $(\ln x_i, \ln y_i)$.

x_i	y_i	$\ln x_i$	$\ln y_i$	$(\ln x_i)^2$	$\ln x_i \ln y_i$
0.5	0.7	-0.693147	-0.356675	0.480453	0.247228
1.0	1.1	0.000000	0.095310	0.000000	0.000000
1.5	1.6	0.405465	0.470004	0.164402	0.190570
2.5	2.1	0.916291	0.741937	0.839589	0.679830
3.0	2.3	1.098612	0.832909	1.206949	0.915044
		1.727221	1.783485	2.691393	2.032673

Example cont.

The corresponding Gaussian normal equations are:

$$\begin{aligned} 2.691393A + 1.727221B &= 2.032673 \\ 1.727221A + 5B &= 1.783485. \end{aligned}$$

Its solution is $A = 0.676257$, $B = 0.123088$, and hence the original parameters are $a = A = 0.676257$ and $b = e^B = e^{0.123088} = 1.130984$, hence the solution is

$$y = 1.130984x^{0.676257}.$$

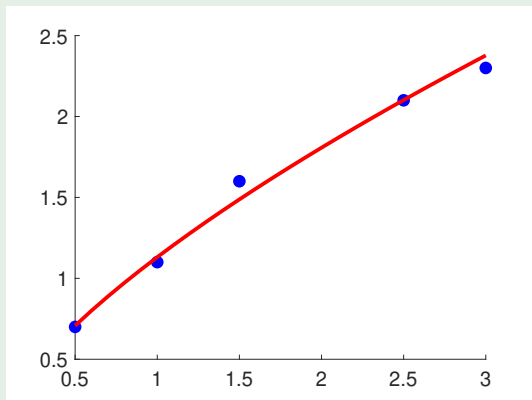
The error of the linear fitting is

$$\sum_{i=0}^4 (0.676257 \ln x_i + 0.123088 - \ln y_i)^2 = 0.007279,$$

and the error of the original nonlinear fitting is

$$\sum_{i=0}^4 (1.130984x_i^{0.676257} - y_i)^2 = 0.019616.$$

Example cont.



Fitting of a power function: $1.130984x^{0.676257}$